

Sioux Falls VA Antibiogram

Calendar Year 2025

Located under Tools tab in CPRS

GRAM NEGATIVE RODS % Susceptible (# tested)

ENTERICS and PSEUDOMONAS	Ampicillin	Ampicillin/ Sulbactam	Cefazolin	Ceftazidime	Ceftriaxone	Cefepime	Piperacillin/ Tazobactam	Meropenem	Ertapenem	Ciprofloxacin	Nitrofurantoin	Gentamicin	Trim/Sulfa
<i>Enterobacter cloacae</i> (23)*				83		96	80	100	96	87	39	100	87
<i>Escherichia coli</i> (180)	63	72	78	90	90	94	98	98	99	84	96	94	85
<i>Klebsiella pneumoniae</i> (47)	0	89	92	96	94	96	98	94	94	91	40	100	89
<i>Proteus mirabilis</i> (40)	75	80	68	98	95	95	100		100	75	0	100	78
<i>Pseudomonas aeruginosa</i> (39)				100		100	92	87		82		100	

RESPIRATORY PATHOGENS % Susceptible (# tested)

<i>Streptococcus pneumoniae</i> (20)* 2024 & 2025 data	Penicillin –parenteral (non-meningitis)			Ceftriaxone			Erythromycin	Trim/Sulf	Doxycycline
	S	I	R	S	I	R	S	S	S
	100	0	0	100	0	0	55	60	60

GRAM POSITIVE COCCI % Susceptible (# tested)

STAPHYLOCOCCUS	Nafcillin	Clinda	Doxy	Vanco	Trim/Sulf	Rifampin
<i>Staphylococcus aureus</i> (142) 26% MRSA 74% MSSA	74	80	94	100	94	100
Coagulase negative <i>Staphylococcus</i> (110)	72	73	96	100		

ENTEROCOCCUS	Ampicillin	Vanco	Nitrofurantoin	Gent Synergy**	Linezolid
<i>Enterococcus faecalis</i> (78)	99	100	100	87	99
<i>Enterococcus faecium</i> (5)* 2024 & 2025 data	20	80	0		100

*Less than 30 isolates may influence the statistical significance of this data.

**Susceptible predicts synergy between aminoglycosides and beta lactams / vancomycin

**Department of
Veterans Affairs**

Memorandum

Date: 1/14/26

From: Pathology and Laboratory Medicine Service

Subj: ANTIBIOGRAM 2025 – VAHCS, Sioux Falls, SD

To: Staff Physicians and Other Health Care Providers

1. **ANTIMICROBIAL STEWARDSHIP** consists of the systematic measurement and timely coordinated interventions designed to promote the optimal use of antimicrobial agents in a hospital system. Since 2017, the Joint Commission has required that all hospitals and nursing care centers have antimicrobial stewardship programs. The Antimicrobial Stewardship Program (ASP) at the SFVAHCS oversees the choice, dosing, route, and duration of administration to optimize clinical outcomes while minimizing unintended consequences. Unintended consequences include antimicrobial resistance and the emergence of pathogenic organisms such as *Clostridioides difficile*, MRSA, VRE, ESBL, and other multidrug resistant organisms like CRE and the new Carbapenem Resistant *Pseudomonas aeruginosa* (CRPA).
2. **ANTIBIOTIC GUIDELINES CLINICAL DECISION SUPPORT SYSTEM (CDSS)** is available to all clinicians at this facility. It can be found within several menus under the Orders Tab. We encourage clinicians to use the guidelines to select empiric therapy and Quick Order their antibiotics. By guiding the use of antimicrobials this facility's ASP has documented decreased rates of resistant pathogens and Hospital Acquired Infections (HAI), as well as improved patient safety and well-being.
3. The **ANTIBIOGRAM** attached to this memorandum includes clinically significant organisms isolated at the facility between January 1st and December 31st, 2025. The data can be relied upon by clinicians to direct empiric antimicrobial therapy prior to receiving the results of susceptibility testing from microbiology. Only organisms causing clinical infections at this facility are included in the **ANTIBIOGRAM**. The **ANTIBIOGRAM** can be found in CPRS under the Tools Bar.
4. Please be aware that a wider variety of antimicrobials are tested than are reported on the culture report from microbiology. Should a patient present a therapeutic problem, consult the Infectious Disease Service or the ID Clinical Pharmacist for real time assistance.
5. **HAND HYGIENE** and **STANDARD PRECAUTIONS** are the backbone of infection prevention and should be observed for ALL interactions with patients.

ANTIBIOGRAM 2025 REPORT SUMMARY:

- **VRE (Vancomycin Resistant Enterococcus):** There was one infection secondary to VRE in 2025, zero in 2024, and five in 2023. Clinically significant VRE cases and isolates remain rare at this facility.
- **MRSA (Methicillin Resistant *Staphylococcus aureus*):** MRSA isolates accounted for 26% of all *S. aureus* isolates in 2025, 25% in 2024, and 25% in 2023. Both MSSA (Methicillin Sensitive *S. aureus*) and MRSA continue to show a high degree of sensitivity to Trimethoprim/sulfamethoxazole (94% susceptible) and Doxycycline (94% susceptible) making them ideal oral options for treatment of staphylococcal infections. One MRSA case was documented in the facility in 2025.
- ***Streptococcus pneumoniae*:** The data indicates 100% are susceptible to penicillin and 100% susceptible to 2nd and 3rd generation Cephalosporins, which are the most appropriate agents for empiric use.

- **ESBL (Extended Spectrum Beta Lactamase)** producing *E. coli*, *Proteus*, and *Klebsiella spp.* are screened and confirmed in our lab. There were eleven new ESBL infections confirmed in 2025, ten in 2024, and fifteen in 2023. In 2025, six of the ESBLs were *E. coli*, two were *K. pneumoniae*, one was *K. oxytoca* and two were *Proteus mirabilis*. To date trends remain stable.
- **CRE (Carbapenem Resistant Enterobacteriaceae)** *E. coli*, *Klebsiella spp.* and others, frequently already ESBL producers, often gain further resistance to Carbapenems under antibiotic pressure. One CRE organism was isolated at this facility in 2025, five in 2024, and one in 2023.
- ***Escherichia coli*:** Nitrofurantoin susceptibility remains high at 96% and is the best first line agent for treatment of lower urinary tract infections like acute cystitis in our population. TMP/SMX and Ciprofloxacin have similar susceptibilities (85% and 84%). When equally efficacious and narrower spectrum options are available, we recommend selecting TMP/SMX over fluoroquinolones.
- ***Pseudomonas aeruginosa*:** Piperacillin/tazobactam is the primary anti-pseudomonal IV antibiotic in use at this facility. Piperacillin/tazobactam susceptibility remains excellent at 92%. Ciprofloxacin susceptibility has slightly decreased to 82%. Two carbapenem resistant *P. aeruginosa* (CRPA) were isolated at our facility in 2025, two in 2024, six in 2023.

Karlene Reid M.D.
Antimicrobial Stewardship Program

Jerry San Diego, M.D.
Clinical Chief, P&LMS

Kelli Jacobsen, MBA, MLS (ASCP)^{CM}
P&LMS Manager

Andrea Aylward, Pharm D.
Antimicrobial Stewardship Program

Elizabeth Watson MLS (ASCP)
Clinical Microbiologist

Nancy Parr, RN, BA
Infection Preventionist